GrantLens: An AI-Powered Research Grant Compliance Auditing & Proposal Evaluation Platform

1. Introduction

Applying for research grants is one of the hardest tasks for university professors. Writing a proposal takes over a hundred hours, yet funding agencies enforce 50-to-80-page rulebooks with strict requirements. Up to 30% of submissions are rejected right away during administrative screening for minor compliance mistakes before reviewers even read the science. Meanwhile, university research offices struggle to manually check hundreds of pages against these rules. Faculty members frequently balance heavy teaching loads and laboratory supervision alongside intense grant deadlines, leaving little room for exhaustive manual audits. Overlooked formatting errors, misaligned figures, and missing disclosures routinely derail promising research initiatives before peer evaluation begins. Traditional writing tools and spell checkers completely fail to detect structural guideline violations or regulatory gaps.

GrantLens solves this by providing an AI assistant that reads both the official rulebook and the draft proposal, automatically catching missing sections, budget mismatches, and scoring errors before submission. It delivers an automated pre-flight audit pipeline that protects research funding investments and elevates institutional submission quality.

2. Problem Statement

- **Administrative Disqualification:** Missing mandatory sections (such as Data Management Plans) causes instant rejection during initial screening.
- **Constraint Violations:** Proposals frequently break hard limits on total page count, section word counts, and maximum funding amounts.
- **Internal Inconsistencies:** Written justifications contradict attached financial tables (e.g., claiming two postdocs in the text while only budgeting for one).
- **Missing Evaluation Benchmarks:** Applicants have no automated mechanism to check if their drafts satisfy official agency grading rubrics prior to submission.
- **Administrative Bottlenecks:** University research offices spend excessive hours manually verifying proposals against fluctuating agency guidelines.

3. Objectives

- **Centralized Document Ingestion:** Develop an automated pipeline to ingest multi-page funding guideline PDFs and proposal drafts.
- **Automated Rule Extraction:** Extract regulatory constraints, mandatory sections, and scoring rubrics using semantic indexing.

- **Compliance Verification:** Build dual-layer validation to detect missing required sections and physical formatting breaches.
- **Cross-Document Reconciliation:** Detect numerical discrepancies between text narratives and structured budget sheets.
- **Rubric-Aligned Scoring:** Generate objective proposal scores (0–100) based on official agency review criteria.
- **Targeted Rewrite Guidance:** Provide actionable, paragraph-level text recommendations to resolve identified gaps.
- **Institutional Dashboards:** Supply university research administrators with real-time compliance oversight and risk analytics.
- **Grounded Regulatory Q&A:** Implement a retrieval-augmented conversational interface enabling researchers to query complex rulebook clauses with citation backing.
- **Audit Report Generation:** Produce standardized, downloadable pre-flight audit summary reports for departmental sign-off and administrative record-keeping.

4. Proposed Solution

Researcher Portal

- Allows applicants to upload funding call PDFs and draft proposals.
- Executes instant pre-submission compliance and constraint scans.
- Displays color-coded risk flags (Red for fatal errors, Yellow for warnings, Green for passed checks).
- Generates cross-document budget reconciliation reports.
- Provides simulated rubric evaluations and targeted rewrite advice.
- Exports pre-flight PDF audit reports for institutional clearance.

Reviewer & Research Office Portal

- Aggregates submitted departmental drafts alongside active funding calls.
- Displays AI-generated executive summaries and compliance scorecards.
- Identifies regulatory risks before external agency submission.
- Enables staff to approve, flag, or request modifications from faculty.

Admin Portal

- Manages user access, departmental privileges, and role authentication.
- Maintains guideline archives and standard funding call templates.
- Oversees vector database maintenance and system audit logs.

5. AI Features

- **Automated Guideline Ingestion:** Uses document layout analysis to segment dense rulebooks into indexed rule chunks.
- **Deterministic & Semantic Verification:** Evaluates physical constraints (page caps) and assesses semantic completeness of required disclosures.
- **Budget vs. Narrative Reconciliation:** Uses Information Extraction to extract personnel, equipment, and costs from narrative text and cross-references them against budget cells.

- **Simulated Peer-Review Scoring:** Evaluates drafts against published evaluation criteria (e.g., Intellectual Merit, Broader Impacts) on a 0–100 scale.
- **Constructive Rewrite Suggestions:** Pinpoints weak or unsubstantiated arguments and generates grounded revision alternatives.
- **Grounded RAG Query Engine:** Retrieves direct answers to specific regulatory questions directly from the indexed guideline text.

6. Methodology

- Requirement Analysis
- System Design
- Database Design
- Frontend Development
- Backend API Development
- AI Module Development
- Integration
- Testing
- Deployment
- Evaluation

7. Proposed Technology Stack

- **Frontend:** React.js (Vite), Tailwind CSS, Recharts
- **Backend:** Django, Django REST Framework (DRF)
- **Document Extraction:** PyMuPDF (fitz), pdfplumber
- **Database & Storage:** PostgreSQL, ChromaDB
- **AI & NLP:** Python, LangChain, Google Gemini API
- **Authentication:** SimpleJWT, Django Authentication
- **Version Control & Deployment:** Git, GitHub, Docker, Render / Railway

8. Expected Outcomes

- **Elimination of Administrative Rejections:** Prevents immediate proposal disqualifications caused by non-scientific formatting or missing sections.
- **Time Savings for Researchers:** Reduces dozens of hours spent manually cross-checking proposals against lengthy guideline documents.
- **Error-Free Financial Alignment:** Catches hidden discrepancies between written technical plans and itemized budget figures prior to submission.
- **Improved Proposal Competitiveness:** Provides objective rubric benchmarking to help applicants strengthen low-scoring sections.
- **Reduced Administrative Overhead:** Streamlines departmental review cycles for university research administration offices.

9. Scope of the Project (Future Enhancements)

- **ERP & Portal Integration:** Connect with university systems and official government submission portals.
- **Literature Verification:** Cross-check citations and novelty against scholarly databases like Semantic Scholar.
- **Live Collaborative Editor:** Provide bidirectional LaTeX and Markdown editing with automated diffs.
- **Multi-Institute Workflows:** Enable joint authoring, cross-university budget splitting, and sign-offs.
- **Mobile Alerts:** Offer companion apps for deadline tracking and chair approval notifications.

10. Conclusion

GrantLens modernizes grant preparation by combining NLP and document intelligence into an easy-to-use platform. It acts as a 24/7 pre-submission reviewer, automatically catching fatal rule omissions, fixing budget contradictions, and grading proposals against official criteria. Beyond individual proposals, the platform transforms institutional research administration into an agile, data-informed operation. It equips faculty with objective, early-stage feedback that clarifies complex regulatory expectations before external review panels convene.

Ultimately, GrantLens positions academic institutions to secure greater research funding by ensuring that no groundbreaking scientific idea is sidelined by bureaucratic oversight.